Graph Theory

Lecture 3

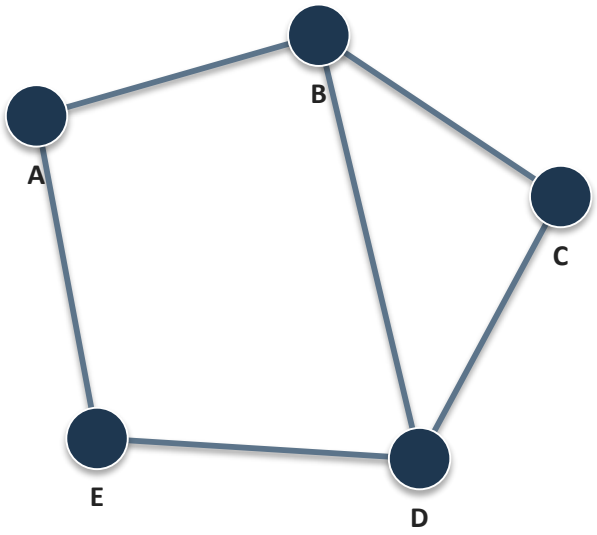
Graph Isomorphism & Structural Vulnerability

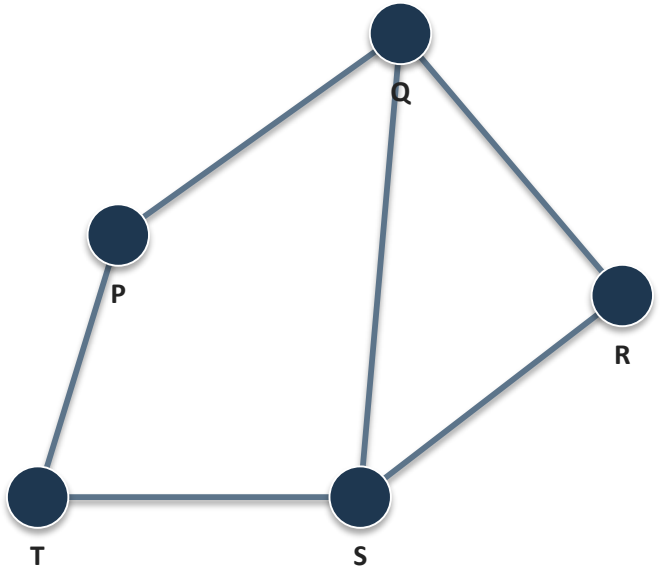
Learning Outcomes

By the end of this lecture, students should be able to:

- Use structural invariants to test whether two graphs can be isomorphic.
- Construct and verify an isomorphism mapping.
- Explain why equal degree sequences are not sufficient for isomorphism.
- Identify cut vertices and bridges in connected graphs.
- Relate cut vertices and bridges to single points of failure in real networks.

Real-Life Motivation: Same Network, Different Drawing





Different labels + different drawing ≠ different structure

Graph Isomorphism: The Core Idea

$$f : V(G) \to V(H) \text{ must be a bijection}$$

$$uv \in E(G) \iff f(u)f(v) \in E(H)$$

The essential requirement is simple: adjacency must be preserved.

Vertex names and positions in the drawing do not matter.

Example mapping from the previous slide: A→P, B→Q, C→R, D→S, E→T.

Structural Invariants: Graph Fingerprints

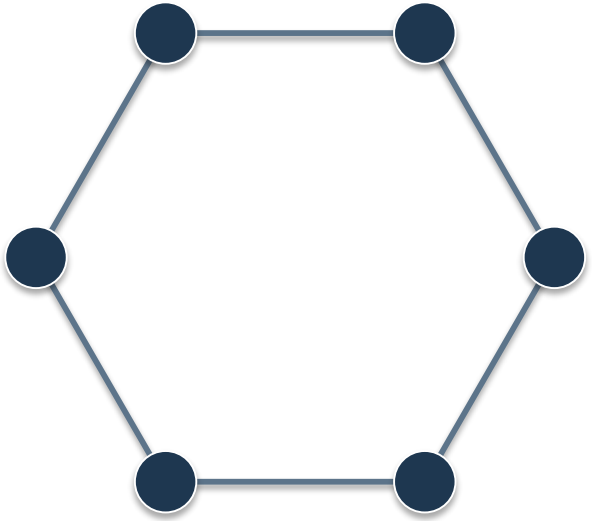
Isomorphic graphs must have the same:

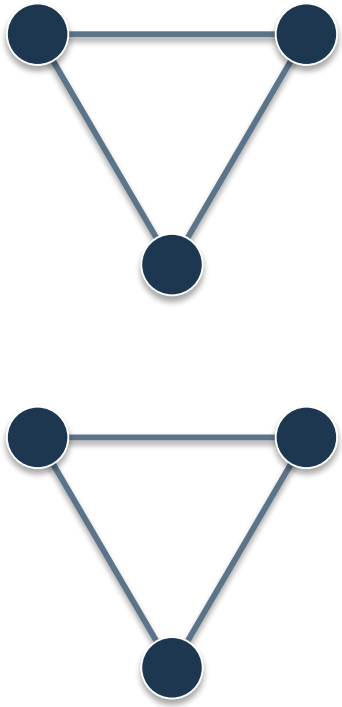
- number of vertices and edges
- degree sequence
- number of connected components
- isolated and pendant vertices
- cycle / triangle structure
- cut vertices and bridges

Different invariant ⇒ NOT isomorphic

Same invariants ⇏ necessarily isomorphic

Same Degree Sequence Does Not Guarantee Isomorphism





C_6: degree sequence (2,2,2,2,2,2), connected

$2K_3$: same degree sequence, disconnected

A Practical Isomorphism-Checking Strategy

1. Compare order $|V|$ and size $|E|$.
2. Compare degree sequences.
3. Match structurally distinctive vertices.
4. Compare neighborhoods, cycles and other invariants.
5. Propose a vertex correspondence.
6. Verify that every adjacency is preserved.

Stop as soon as one invariant differs.

Real-Life Connection: Molecular Structure

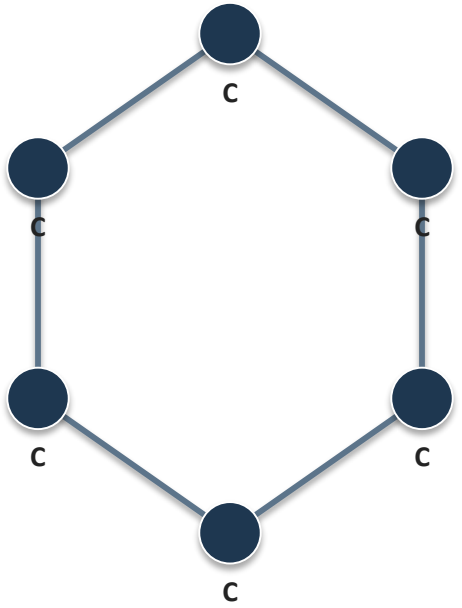
A molecule can be represented as a graph:

- atoms → vertices
- chemical bonds → edges

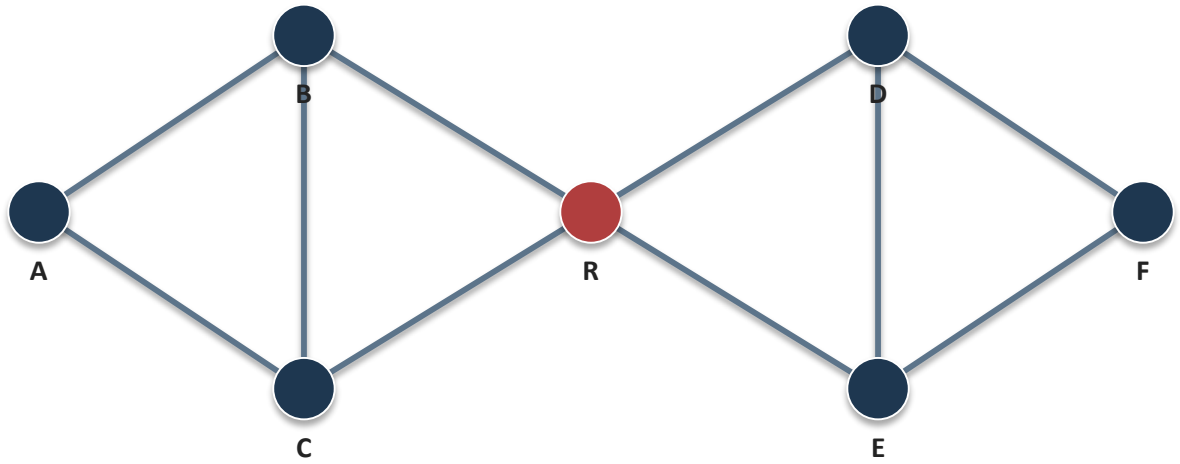
Two drawings may look different because of orientation or layout.

The structural question is whether the same atoms have the same bonding relationships.

Graph matching/isomorphism ideas therefore arise naturally in chemical structure comparison.



New Question: How Vulnerable Is the Structure?



If router R fails, the network separates into two parts.

Cut Vertex / Articulation Point

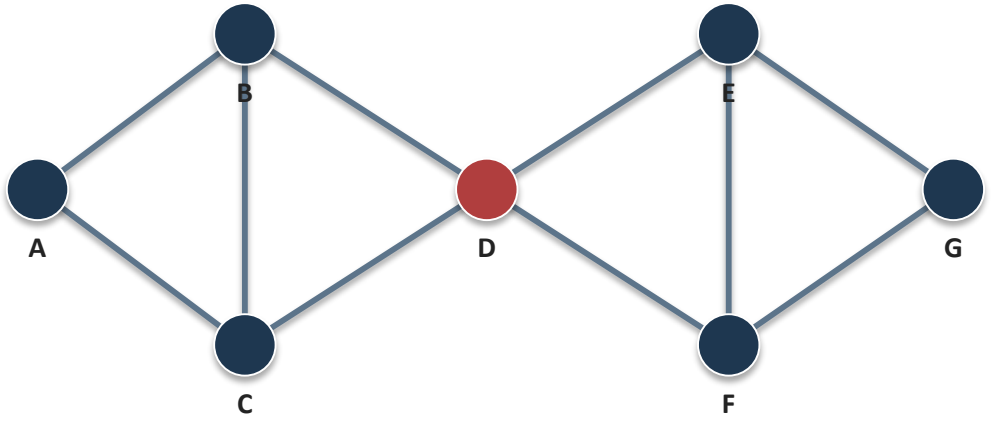
A vertex v is a cut vertex if removing v (and its incident edges) increases the number of connected components.

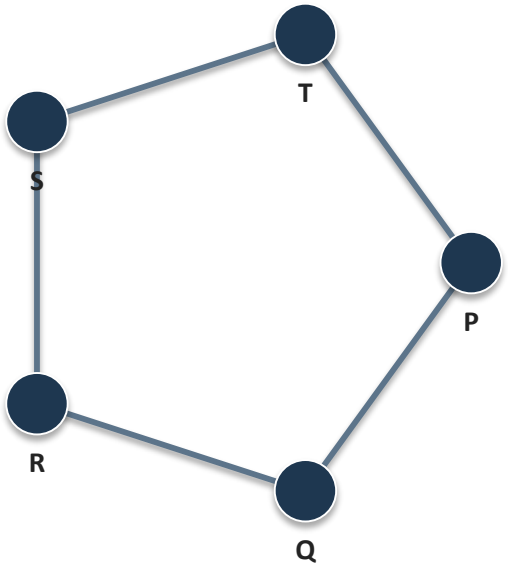
Real-life interpretation:

- critical router or switch
- airport connecting regions
- road junction
- key communication hub

Cut vertex = vertex-based single point of failure

Example: Which Vertex Is Critical?





Left: D is a cut vertex

Right: C_5 has no cut vertex

Bridge / Cut Edge

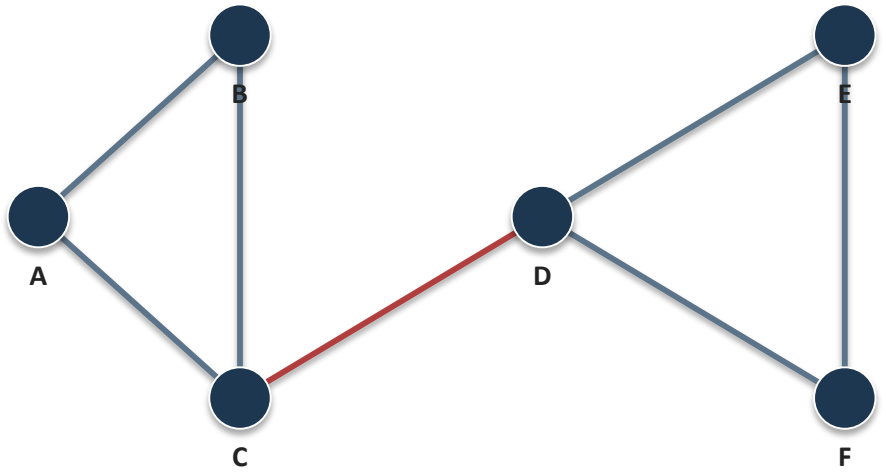
An edge e is a bridge if removing it increases the number of connected components.

Real-life examples:

- a single road connecting a village to the main road network
- a fiber link joining two network segments
- a communication cable with no alternative route

Bridge = edge-based single point of failure

Pictorial Example: Finding a Bridge



Edge C–D is the only connection between the two cycles.

Remove C–D → graph becomes disconnected → C–D is a bridge.

Key Result: Bridges and Cycles

e is a bridge ⇔ e belongs to no cycle

Why?

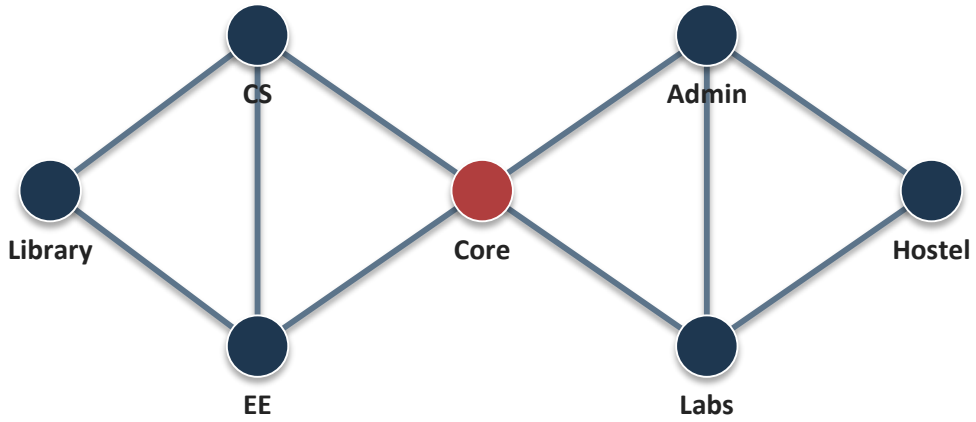
If e lies on a cycle, there is another route between its endpoints.

Removing e therefore does not destroy connectivity between those endpoints.

If e lies on no cycle, there is no alternate path connecting its endpoints.

Network interpretation: cycles provide redundancy.

Real-Life Case: University Fiber Network



Suppose vertices are buildings/routers and edges are fiber links.

Questions:

- Which router is a single point of failure?
- Are there any bridge links?
- Where could one additional link be added to improve resilience?

Design Thinking: Remove the Single Point of Failure

Graph theory is not only about detecting weaknesses — it can guide redesign.

Adding an alternative connection can:

- place a critical edge on a cycle,
- remove bridge status,
- create an alternate route during failure.

Engineering interpretation: redundancy usually increases resilience, but may increase cost.

Next week we quantify this trade-off using vertex and edge connectivity.

Concept Check

1. If two graphs have the same degree sequence, must they be isomorphic?
2. Can isomorphic graphs have different numbers of cut vertices?
3. Can an edge on a cycle be a bridge?
4. Does every nontrivial tree contain a bridge?
5. Why are cycles useful in communication networks?
6. Can a connected graph have neither cut vertices nor bridges? Give an example.

Lecture Summary

Graph isomorphism asks whether two graphs have the same underlying structure.

Structural invariants help reject non-isomorphic graphs efficiently.

Equal degree sequences alone do not prove isomorphism.

Cut vertices are critical vertices whose failure disconnects a graph.

Bridges are critical edges whose failure disconnects a graph.

Cycles create alternative routes and therefore structural redundancy.

Next: Vertex Connectivity, Edge Connectivity, Disjoint Paths & Menger's Theorem.